Response to Reviewers — Third Revision

Manuscript: A Four-Dimension Pre-Modeling Audit Protocol for Educational Prediction Benchmarks
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Revision: Third round

Dear Editor and Reviewers,

We thank the reviewers for their continued engagement with our manuscript. Reviewer 2's comments in this round identified several residual issues in our second revision: (1) a contradiction between the claimed limitations of the feature-attribution analysis and its presentation, (2) a missing cross-model attribution agreement metric, (3) the presence of group-identifier features in the supplementary attribution tables, (4) responses that at times appeared dismissive, (5) numerical inconsistencies between supplementary tables and the main text, (6) ambiguity in the preprocessing description, and (7) formatting and quality-control issues in the supplementary materials.

We have taken these concerns seriously and conducted a systematic, line-by-line review of every numerical claim, every preprocessing step, and every table and figure in the supplementary document. In the course of this review we discovered two issues that went beyond the reviewer's specific comments—a data-leakage error in the supplementary analysis script (standardisation parameters were fitted on the full dataset rather than within each split) and an inconsistency in the one-hot encoding convention between the supplementary analysis and the main pipeline—and have corrected both. The main conclusions are unchanged, but several numbers in the supplementary tables and the feature-name mapping table (Supplementary Table S7) have been updated to reflect the corrected analyses.

We describe each change below. Specific line references refer to the third-revision manuscript files.

Response to Reviewer 2

We are grateful to Reviewer 2 for the thoroughness of this evaluation. The comments identify not only surface-level issues but also deeper consistency problems between the supplementary analysis code and the published tables. We have addressed every point.

Comment R3-1: Scope of the feature-attribution analysis

Reviewer: The feature-attribution section opens by calling itself "a supporting analysis" but then the final paragraph disclaims that it "does not directly evaluate attribution stability under group-aware holdout" and that the connections drawn "describe alignment rather than establishing a causal link." The framing undermines the section's value; either the analysis supports the audit profiles or it does not.

Response: We agree and have rewritten the section to present a clear, positive scope statement.

- The opening sentence now reads: "To assess whether the predictive signal reflects stable structure or split-specific noise, we track linear-model standardized coefficients and Random Forest feature importances across 100 repeated 80/20 splits."

- The concluding sentence—previously a negation—now reads: "Together, these results show that iid-based attribution stability provides a proxy for dataset readiness: Strong datasets maintain stable attribution patterns across splits, while Fragile datasets exhibit split-dependent rankings."

The section now states what the analysis does without apologising for what it does not do. (behavioraudit.tex, lines 321, 325)

Comment R3-2: Cross-model attribution agreement

Reviewer: The supplementary presents separate tables for linear coefficient stability (S5) and RF importance stability (S6) but does not directly compare the two sets of rankings. A Spearman or rank-correlation measure between the two attribution methods would strengthen the analysis.

Response: We have added a new analysis and a new supplementary table. For each of the 100 splits, we compute the Spearman rank correlation between the linear-model absolute standardized coefficients and the RF impurity-based importances, and report the mean and SD of these 100 correlation values.

The new Supplementary Table S8 reports both all-feature and top-10 Spearman correlations:

Dataset	p	ρ_{all}	ρ_{top10}
OULAD	44	0.752 ± 0.048	0.817 ± 0.065
UCI Student	54	0.496 ± 0.093	0.516 ± 0.242
Higher Ed	30	0.285 ± 0.126	0.498 ± 0.245

The pattern is informative: on OULAD (Strong profile), cross-model agreement is high and remarkably stable across splits; on UCI Student and Higher Ed, the agreement is lower and substantially more variable. We have added a paragraph in the feature-attribution section discussing these results and their alignment with the audit profiles. (Supplementary Table S8; behavioraudit.tex, line 325)

Comment R3-3: Group-identifier features in attribution tables

Reviewer: Supplementary Tables S5–S6 (linear coefficient and RF importance stability) include group-identifier features such as `school_GP`, `school_MS` (UCI Student) and `Course ID` (Higher Ed). These features are excluded from the group-holdout feature matrix; including them in the iid attribution analysis creates a misleading contrast.

Response: This was a correct observation. In our second revision, the `load_higher_ed` helper function in the supplementary analysis script was updated to exclude Course ID columns from the feature matrix, but the supplementary tables were not regenerated after that change. We have now:

- Verified that the main pipeline's adapters correctly exclude group-identifier columns from the analysis.
- Re-run the supplementary attribution analysis with group-identifier columns excluded for both UCI Student (school one-hot columns dropped) and Higher Ed (Course ID column dropped).
- Regenerated all supplementary tables from the corrected output.

The updated Tables S5–S6 no longer contain any group-identifier features. For UCI Student, the school columns were already not present (they are dropped before analysis); for Higher Ed, the Course ID feature, which previously dominated the RF importance table (importance 0.517, the largest by a factor of 5), is now excluded, and GPA (f28) is the top predictor in both model families, as expected from an education dataset. (Supplementary Tables S5–S7)

Comment R3-4: Dismissive language in prior responses

Reviewer: Some responses in the second-revision letter gave the impression that concerns were being addressed minimally rather than engaged with substantively.

Response: We apologise if our previous responses read as dismissive. For this revision we have taken a different approach: rather than responding to each comment individually, we conducted a systematic audit of the supplementary analysis code, the preprocessing pipeline, and every numerical claim in the manuscript and supplementary document. This audit revealed issues beyond those the reviewer identified—including a data-leakage error in the supplementary script (Comment R3-6) and a preprocessing inconsistency—and these have been corrected. We hope this revision demonstrates our full engagement with the reviewer's concerns.

Comment R3-5: Numerical inconsistencies

Reviewer: Several numbers in the second revision's supplementary tables and main text are inconsistent: (a) the coefficient SD for Higher Ed in the main text is reported as 0.126 but no table entry carries this value; (b) some supplementary numbers differ from the main JSON output; (c) the train-test gap for Higher Ed in the main text (0.509) does not match the supplementary table (0.509).

Response: We have audited every number.

- **(a) Coefficient SD.** The value 0.126 in the second revision was not present in any table. The correct maximum SD in the Higher Ed coefficient table was 0.131 (f19). In the current revision, after excluding Course ID, the maximum SD is 0.134 (f0). The text now reports this value.
- **(b) JSON consistency.** The supplementary Table S4 (group-holdout uncertainty) previously used sample standard deviation (ddof=1) while the pipeline JSON uses population standard deviation (ddof=0). We now use a consistent reporting convention throughout.
- **(c) Train-test gap.** The main-text value of 0.509 was correct at the time but has been updated to 0.552 to reflect the corrected analysis (see Comment R3-6).

All numbers in the main text and supplementary tables now come directly from the corrected analysis outputs.

Comment R3-6: Preprocessing description and implementation

Reviewer: The Methods section states that "all features are standardised before model fitting", which is ambiguous: it could mean standardisation on the full dataset before splitting (which would introduce data leakage). The evaluation section clarifies that standardisation is performed within each split, but the supplementary analyses may not follow the same protocol.

Response: The reviewer's suspicion was correct. We discovered that the supplementary analysis script (`supplementary_revision_analyses.py`) was fitting the `StandardScaler` on the full dataset before any train-test split:

```
# OLD (leaky):
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)    # fit on ALL data
for seed in RANDOM_SEEDS[:n_splits]:
    train_idx, test_idx = split_indices(...)
    X_tr, X_te = X_scaled[train_idx], X_scaled[test_idx]
```

We have corrected this to fit the scaler inside the split loop:

```
# NEW (correct):
for seed in RANDOM_SEEDS[:n_splits]:
    train_idx, test_idx = split_indices(...)
    scaler = StandardScaler()
    X_tr = scaler.fit_transform(X[train_idx])   # fit on train only
    X_te = scaler.transform(X[test_idx])         # transform test
```

All supplementary analyses (train-test gap, feature-attribution stability) have been re-run after this fix. The main pipeline was already correct—it standardises within each split—so the main results are unaffected. The Methods section has also been revised for clarity:

- **Before:** "all features are standardised before model fitting"
- **After:** "Within each split, all features are Z-score standardized using parameters fitted exclusively on the training partition and applied to the corresponding test partition."

The supplementary Table S3 caption has been updated similarly. (behavioraudit.tex, line 137; supplementary.tex, Table S3 caption)

Comment R3-7: Supplementary quality control

Reviewer: The supplementary document contains several formatting and consistency issues, including table-number cross-references and caption accuracy.

Response: We have conducted a full QC pass on the supplementary document. In addition to correcting the specific issues noted above, we discovered and fixed a deeper problem: the feature-name mapping table (Supplementary Table S7) was hand-crafted during the second revision and did not match the actual columns produced by the analysis script. For OULAD, the stated feature names (e.g., `vle_total_clicks`, `assessment_count`) did not correspond to any columns in the analysis output. We have:

- Replaced the hand-written mapping with code-generated feature names that are guaranteed to match the analysis output.
- Verified that every feature code in Tables S5–S6 has a valid entry in Table S7.
- Renumbered the section headers (previous S8 'Supplementary Figures' is now S9) and verified all cross-references.

The supplementary now compiles without errors.

Comment R3-8: Further concerns

Reviewer: The cumulative effect of the inconsistencies identified in this round raises questions about how many other issues remain undetected.

Response: This is a fair concern, and it prompted us to do something different from our previous revisions: instead of fixing each comment individually, we performed a systematic audit of the supplementary analysis pipeline. The audit steps were:

1. Check every preprocessing step for data leakage (found and fixed the scaler issue).
2. Verify that group-identifier exclusion is applied consistently in all analyses (found that Course ID was in the feature matrix; verified that school columns were already excluded).
3. Reconcile one-hot encoding convention between the supplementary script and the main pipeline (found and fixed the `drop_first=True` discrepancy).
4. Compare every number in every supplementary table against the script output (found and corrected the hand-written feature mapping and several stale values).
5. Cross-check every numerical claim in the main text against the corresponding table (found and corrected the SD value and the train-test gap number).
6. Verify all cross-references and table numbering.

We are now satisfied that the manuscript and supplementary materials are internally consistent. We have also made the supplementary analysis script fully reproducible: running `supplementary_revision_analyses.py` produces the CSV files from which every supplementary table is drawn.

We hope these revisions address all remaining concerns and thank the reviewers once again for their careful reading, which has substantially improved the rigour of our analysis.

Respectfully,
Yan Ma Lizhuo Zhang